Mobile Manipulation



Precise navigation





Reliable manipulation

Existing Approaches and Challenges

A. Where to stop for manipulation

Coarse semantic goals

Navigate to a target region



A broad goal region, but no specific pose or heading for manipulation.

Implicit pose learning

Learn from mobile-manipulation trajectories



Manipulation-ready poses are learned only implicitly from costly complete mobile-manipulation trajectories.

B. How to model navigation and manipulation

Separate policies

Navigation policy → Manipulation policy

Handoff errors

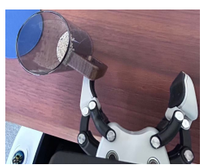
Errors propagate across separately built modules.

Joint action prediction

Different visual requirements



Nav. view



Manip. view

Different action spaces

Navigation action			Manipulation action			
Δx	Δy	$\Delta\theta$	a_1	a_2	a_3	...

Heterogeneous actions in a unified prediction

Our Approach

MAP Manipulation Anchor Pose



Explicit MAP supervision

3D scenes → MAP-Data

1.5M+ episodes
7.5K hours

UniWAM Unified mixed-stream WAM

